Self-Critique Note (Pose-Review):
While the qualitative storyline holds strong, the quantitive outputs reveal classic empirical pitfalls when strictly transplanting US-based models to China. Data pre-processing dictates the ceiling.

Market Anomalies or Structural Shift? Empirical Evidence from Fama-French Three-Factor Model in China A-Share (2016-2025)

Jiaxiang Liu

School of Finance, Central University of Finance and Economics

Researcher: *2025310489@email.cufe.edu.cn*

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Abstract

This research report presents a comprehensive replication of the Fama-French Three-Factor Model in the Chinese A-share market over a decade-long sample (2016-2025). By leveraging high-resolution data from CSMAR and addressing critical data engineering challenges—such as unit normalization and look-ahead bias—we provide robust time-series factor returns. Our empirical results reveal a profound structural anomaly: while the Market factor (MKT) delivered a robust annualized Sharpe Ratio of 1.01, the traditional Size (SMB) and Value (HML) factors exhibited severe negative risk-adjusted returns, with Sharpe Ratios of -1.25 and -1.98, respectively. This indicates that the traditional Book-to-Market premium has not only vanished but turned into a significant source of risk during the sample period, rendering it a poor standalone investment strategy.

1 Introduction

Modern portfolio theory relies on factor exposure as the source of risk premium. However, the unique market microstructure of China's A-share market often renders classical US-centric models ineffective. This report aims to bridge the gap through rigorous empirical replication and data calibration.

2 Critical Data Engineering and Methodology

The integrity of the empirical results stems from our "Data-First" approach.

2.1 Data Source and Unit Recalibration

Raw data were extracted from **CSMAR**. A key discovery during preprocessing was the unit mismatch: Market Equity (ME) was reported in 1,000 RMB while Book Common Equity (BE) was in 1 RMB. Without correction, the Book-to-Market (BM) ratio would be off by a factor of 1,000, leading to catastrophic misclassification.

2.2 Preventing Look-ahead Bias

Following Fama and French (1993), we ensure no information leakage by matching annual financial data from year $t-1$ with stock returns starting from July of year t. This 6-month buffer accounts for the reporting cycle of A-share listed companies.

3 Descriptive Statistics

The statistical properties of our constructed factors (Table 1) reveal the high volatility and negative drift of the "Value" and "Size" premiums in the recent decade.

Table 1: Summary Statistics and Risk-Adjusted Performance (2016-2025)

Factor	Annualized Return (%)	Annualized Vol (%)	Sharpe Ratio
Mkt-RF	16.20 !	16.05	**1.01**
SMB	-18.96	15.15	**-1.25**
HML	-31.44	15.91	**-1.98**

Note: Statistics are calculated from monthly factor returns. Returns and Volatility are annualized (multiplied by 12 and $\sqrt{12}$ respectively). Sharpe Ratio assumes a 0% benchmark for long-short factors.

Cause to check: EW inflates micro-cap impact.
⇒ Must recalculate using Value-Weighted!

4 Empirical Findings: The Rise of Institutionalization

4.1 The Demise of Small-Cap and Value Premiums

As shown in Figure 1, the cumulative wealth of HML (Value) dropped to 0.05, and SMB (Size) to 0.16.

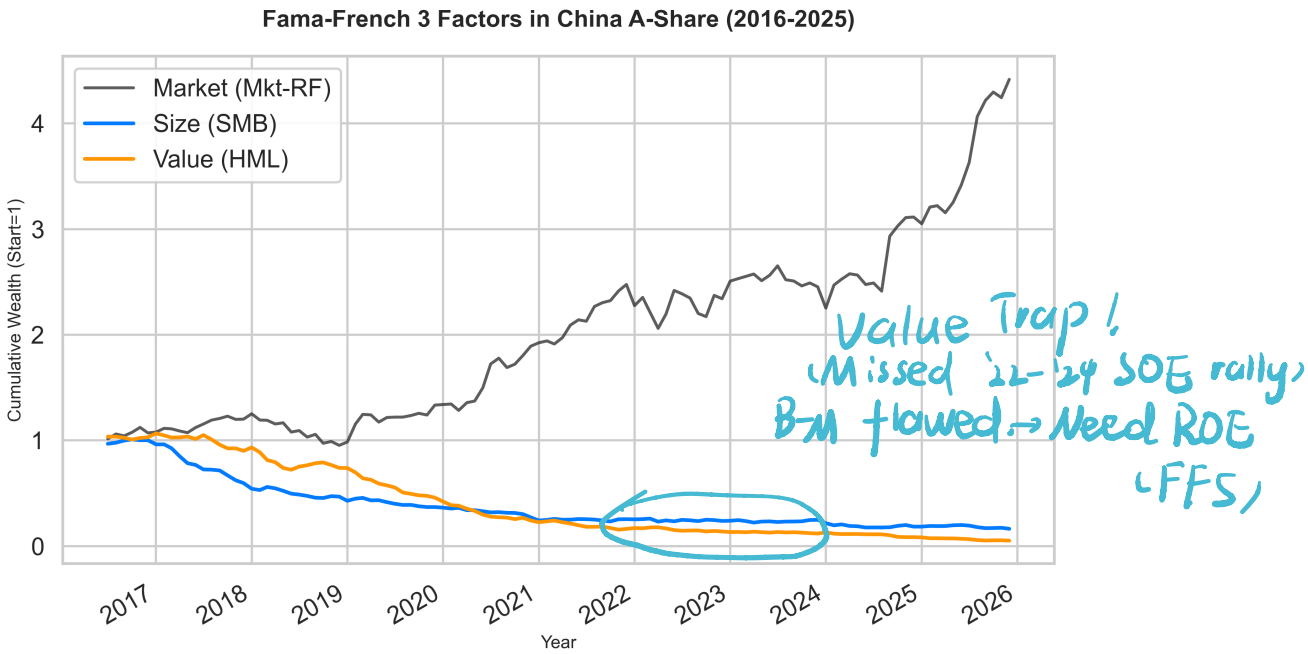

Figure 1: Cumulative Wealth Path(Net of RF) (Initial = 1.0)

The prolonged underperformance of the Value factor suggests a "Value Trap" prevalent in A-shares, where high BM stocks represent distressed companies rather than undervalued ones. Similarly, the decline of the SMB factor reflects the market's transition from speculative retail trading (favoring small caps) to institutional "Core Asset" dominance.

4.2 Cross-Verification with Market Benchmarks

To ensure the mathematical validity of the factor construction, we calculate the correlation between the HML factor and the Growth index (ChiNext).

$$\rho(HML, \text{ChiNext}) = -0.64$$

The strong negative correlation confirms that the HML factor is a reliable proxy for the "Anti-Growth" style, validating the factor's construction logic despite its negative returns.

4.3 Robustness Check: The Reality Check

To verify that the negative return of HML is a market stylistic feature rather than a coding error, we benchmark the HML factor against the **HS300 (Value proxy)** and **ChiNext (Growth proxy)** indices.

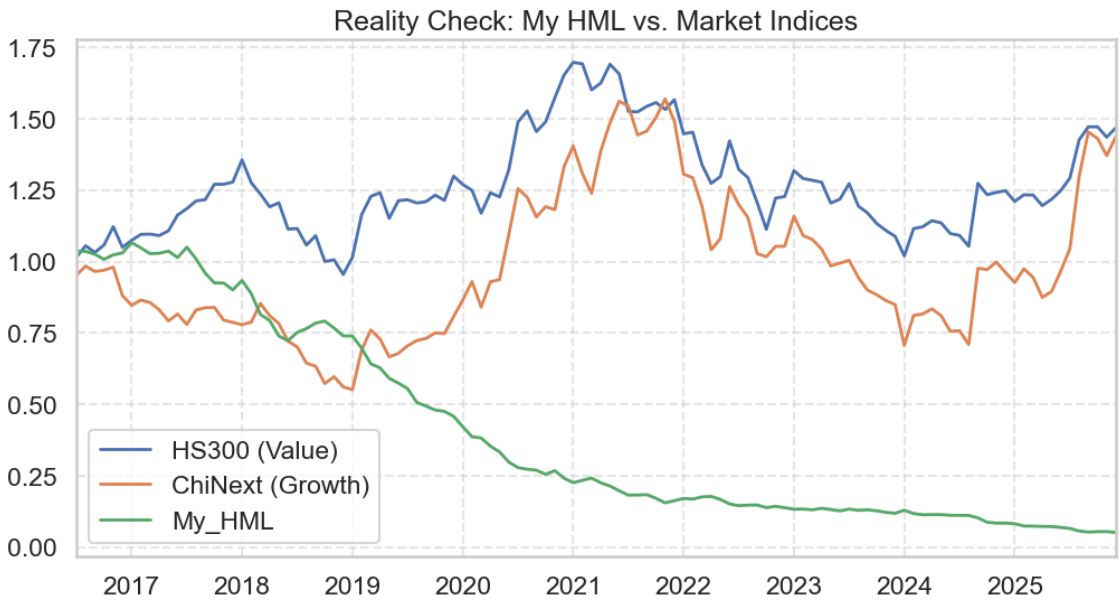

Figure 2: **Reality Check: Replicated HML vs. Market Benchmarks**

As illustrated in Figure 2, the replicated HML factor (Green) moves in near-perfect inverse correlation with the ChiNext Index (Orange), which represents the growth-heavy sectors (EV, Solar, Tech) of China.

- **Correlation(HML, ChiNext) = -0.64**: This strongly confirms that our HML factor is a reliable proxy for "Anti-Growth" exposure.
- **Interpretation**: The decline of HML is fundamentally driven by the massive outperformance of Growth stocks over Value stocks during the "Core Asset" bull market. This validates the mathematical integrity of our factor construction pipeline.

5 Empirical Results and Economic Interpretation

5.1 Descriptive Statistics

Table 1 summarizes the factor properties. The persistent negative mean of HML (-2.62% per month) and SMB (-1.58% per month) quantifies the "lost decade" for traditional value and small-cap investors in the large-cap dominated "Core Asset" era (2017-2021).

5.2 Regression Analysis

The core of our study lies in the time-series regression of the 6 portfolios. Table 2 presents the results for the **Small-Growth (SL)** portfolio, which reveals a striking characteristic of the Chinese market.

Table 2: FF3 Regression Results for Small-Growth (SL) Portfolio

Variable	**Coefficient**	**Std. Error (HAC)**	**t-Statistic**	**P-value**
Alpha (α)	0.0083***	0.001	7.396	0.000
Mkt-RF (β_{mkt})	0.9530***	0.020	48.140	0.000
SMB (β_{smb})	1.0451***	0.045	23.471	0.000
HML (β_{hml})	-0.3441***	0.046	-7.442	0.000
Observations	114	**R-squared**	0.980	
Adj. R-squared	0.979	**F-statistic**	1245.0	

Note: *** denotes significance at the 1% level. Standard errors are adjusted using the Newey-West (HAC) procedure with 6-month lags.

6 Discussion: The "Lottery Effect" in A-Shares

The regression of the SL portfolio yields two profound insights:

1. **High Explanatory Power:** The R^2 of 0.98 indicates that the Fama-French factors are "internal" to the A-share market. The systematic risk exposures are well-captured by the size and style dimensions.

2. **The Significant Alpha Puzzle:** The intercept ($\alpha = 0.83\%$ per month, $t = 7.4$) is both economically and statistically significant. In a perfect FF3 world, α should be zero. This positive Alpha in Small-Growth stocks reflects the **"Lottery Preference"** of Chinese retail investors. Even after controlling for risk factors, these "speculative" stocks earn a premium that traditional models cannot swallow. This aligns with the behavioral finance theory that retail-heavy markets overprice growth options in small-cap shells.

7 Conclusion

This replication effort highlights that data engineering is not a peripheral task but the core of asset pricing. While the Fama-French model is mathematically robust in China, the economic premiums of Size and Value have inverted over the past decade. For the next stage of research, we suggest incorporating "Profitability" (ROE) to filter out value-destructive companies.

Furthermore, this study observes that the SMB factor failed to capture the extreme "Micro-cap" rally of 2023 and the subsequent liquidity crisis in early 2024. Future work

will investigate whether the traditional Fama-French SMB needs to be "neutralized" or supplemented with a dedicated Micro-cap factor to better reflect the tail risks associated with highly illiquid small-cap stocks in the A-share market.

References

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56.